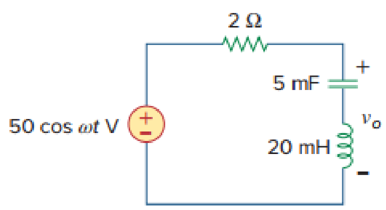
Chapter 9, Problem 34P

Problem

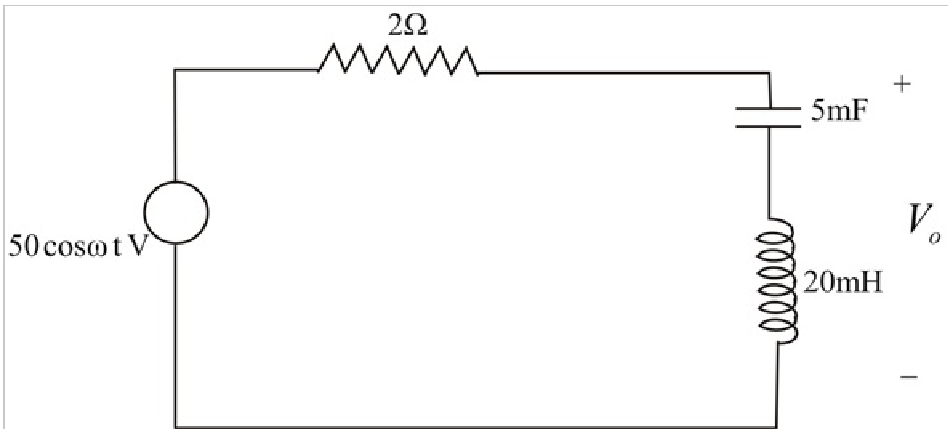
What value of ω will cause the forced response, v_o, in Fig. 9.41 to be zero?

Figure 9.41:



Step-by-step solution

Step 1 of 3



Step 2 of 3

$$V = 50\,Cos\,\omega t = 50\angle 0^\circ$$

$$Z_1 = 2\pi$$

$$Z_2 = \frac{1}{j\omega C} + j\omega L$$

$$= \frac{-j}{\omega\times 5\times 10^{-3}} + j\omega(20\times 10^{-3})$$

$$= \frac{-j}{\omega}0.2\times 10^{+3} + j\omega(20\times 10^{-3})$$

$$V_0 = \frac{Z_2}{Z_1+Z_2}V$$

$$= \frac{\frac{-j}{\omega}(0.2\times 10^3)+j\omega(20\times 10^{-3})}{\frac{-j}{\omega}(0.2\times 10^3)+j\omega(20\times 10^{-3})+2}\{50\angle 0^\circ\}$$

Step 3 of 3

For V0 to be zero,

$$\frac{-j}{\omega}(0.2\times10^{3}) + j\omega(20\times10^{-3}) = 0$$

$$-j(0.2\times10^{3}) + j\omega^{2}(20\times10^{-3}) = 0$$

$$0.2\times10^{3} = \omega^{2}(20\times10^{-3})$$

$$\omega^{2} = \frac{0.2\times10^{3}}{20\times10^{-3}} = 0.01\times10^{6} = 10000$$

$$\omega = 100\ Hz\ .$$